AGGIE HACK 2026 · FAIRLIGHT ADVISORS

Nonprofit Financial Resilience Analytics

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A data-driven platform for Fairlight Advisors to assess, benchmark, and stress-test 130K+ U.S. nonprofits using IRS Form 990 data.

The Problem

?

Where to donate?

Funders guess where to allocate philanthropic capital without data.

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Am I healthy?

Nonprofit leaders can't benchmark themselves against true peers.

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Who needs help?

Capacity builders like Fairlight advise clients without data.

400K+ Form 990s filed. The data is public. The decisions to invest remain guesswork.

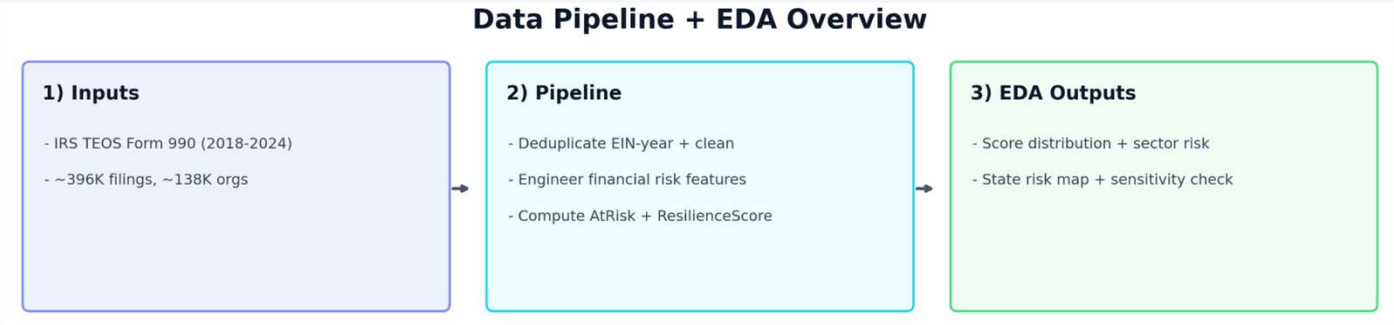
The Data We Started With

Not a sample — the whole sector.

396,910	137,938	7	59
Total filings	Unique organizations	Tax years (2018–2024)	States & territories

Why data quality mattered:

IRS batch deduplication · Negative expense correction · NTEE fallback via mission text

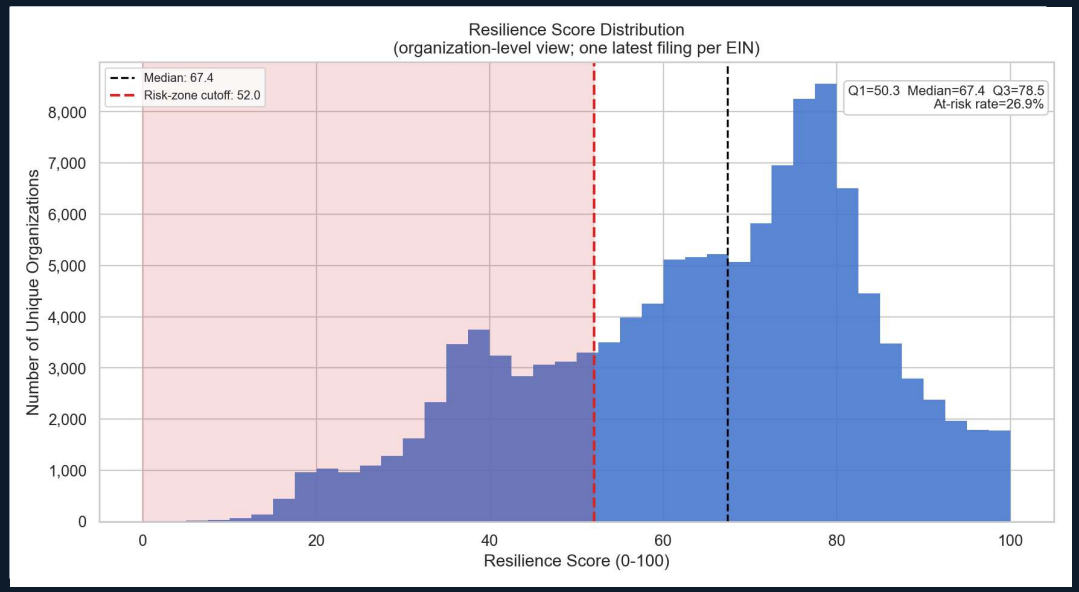


EDA Finding #1

The Sector Is More Fragile Than It Looks

26.9%

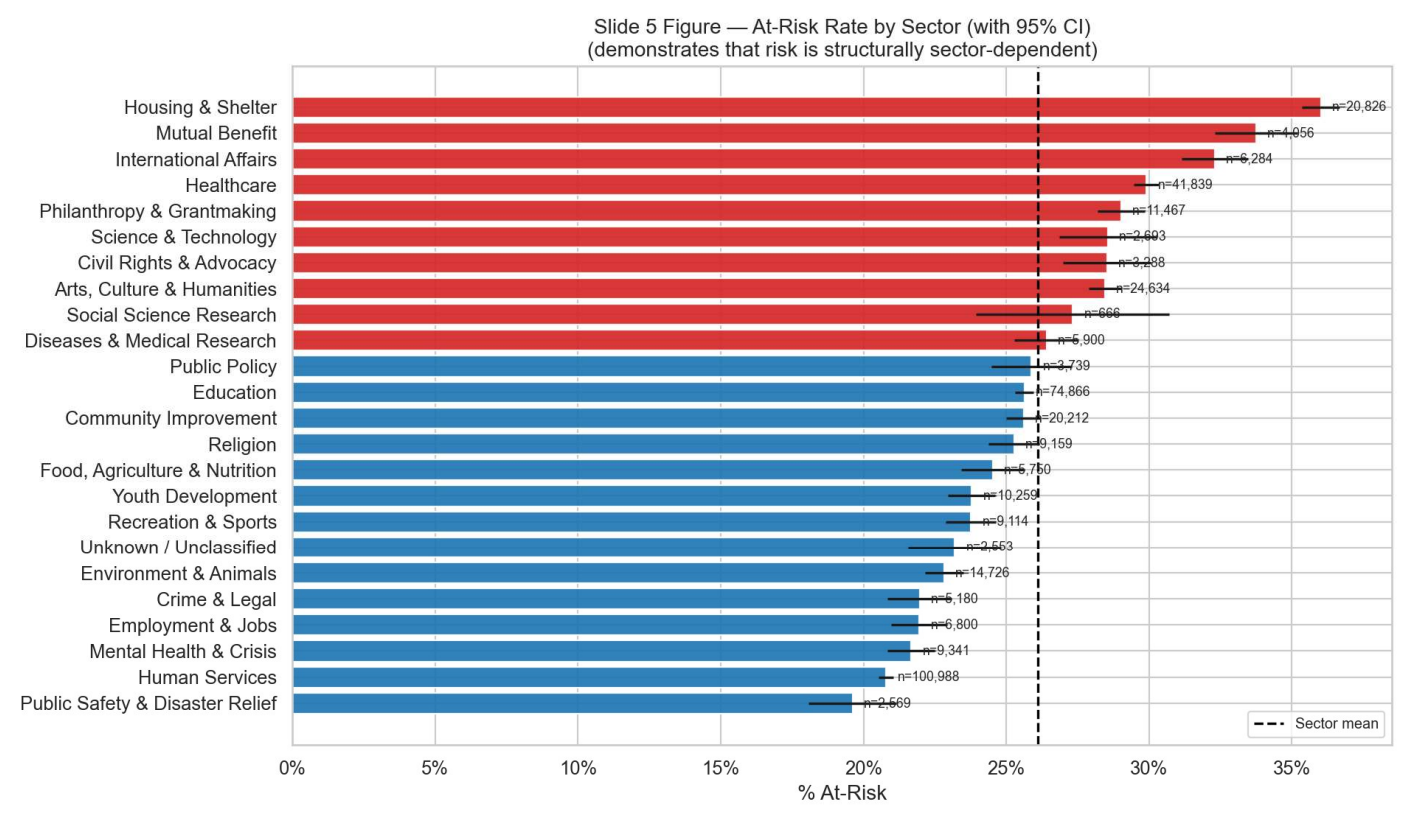
At-risk rate across the
130,000 organizations



The shape of the curve was the first signal this problem was worth solving at scale.

EDA Finding #2

Risk Isn't Random



Housing & Shelter, Mutual Benefit, and International Affairs, consistently score lowest

EDA Finding #3

The Wrong Intuition About Failure

Intuition said low program spending = at risk. The model disagreed.

Debt load and fundraising momentum predict failure more than mission efficiency. Size matters too.

What We Built From These Findings

The Resilience Score

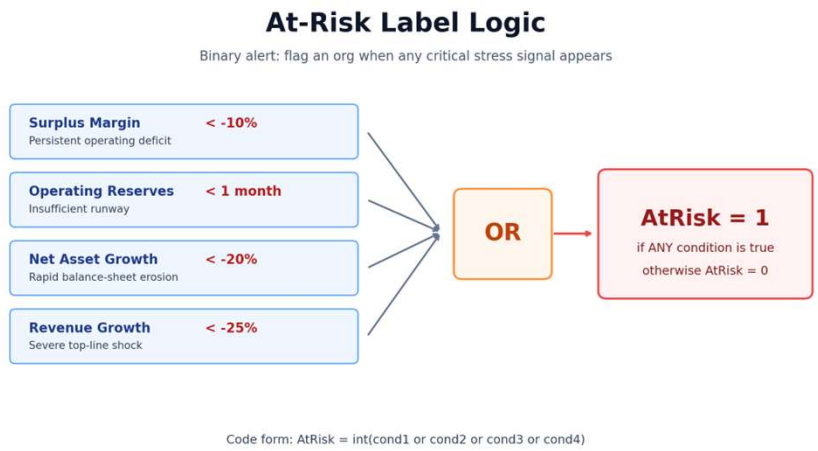
Component	Weight	Why — grounded in EDA findings
Operating Reserves	30 pts	Top stability signal — directly counters low-reserve at-risk definition
Revenue Diversification	20 pts	Counters grant dependency — contribution growth was predictor #2
Program Efficiency	20 pts	Core mission metric — lower weight than expected from EDA
Surplus Margin	15 pts	Deficit is an at-risk trigger — baked into score formula
Low Debt	15 pts	Debt Ratio was the #1 predictor — explicitly in the formula

Strong ≥ 70

Moderate 40–69

At Risk < 40

At-Risk Score



Why this works

- Covers 4 different stress pathways
- Transparent and easy to explain
- Useful as an early warning alert

Component Definitions

- SurplusMargin: Measures whether the nonprofit is consistently earning more than it spends, with negative values signaling operating deficits.
- OperatingReserveMonths: Measures cash runway as months of expenses covered by net assets, with low values meaning weak short-term resilience.
- NetAssetGrowthPct: Measures year-over-year change in net assets, where large declines indicate structural financial deterioration.
- RevenueGrowthPct: Measures year-over-year revenue momentum, where sharp drops indicate funding instability and elevated risk.
- Values: set for transparent early-warning thresholds

We turned these findings into six decision-making tools.

01 Executive Overview

02 Peer Benchmarking

03 Resilience Explorer

04 Stress Test Simulator

05 Hidden Gems Finder

06 Brand Map

Executive Overview

Why we built it

Funders need a sector-wide scan before zooming in. EDA showed risk is unevenly distributed — you need the full picture first.

What it shows

KPI cards (at-risk rate, median reserves), resilience distribution, state choropleth, sector ranking bar chart. Orange alert fires when >20% at risk.

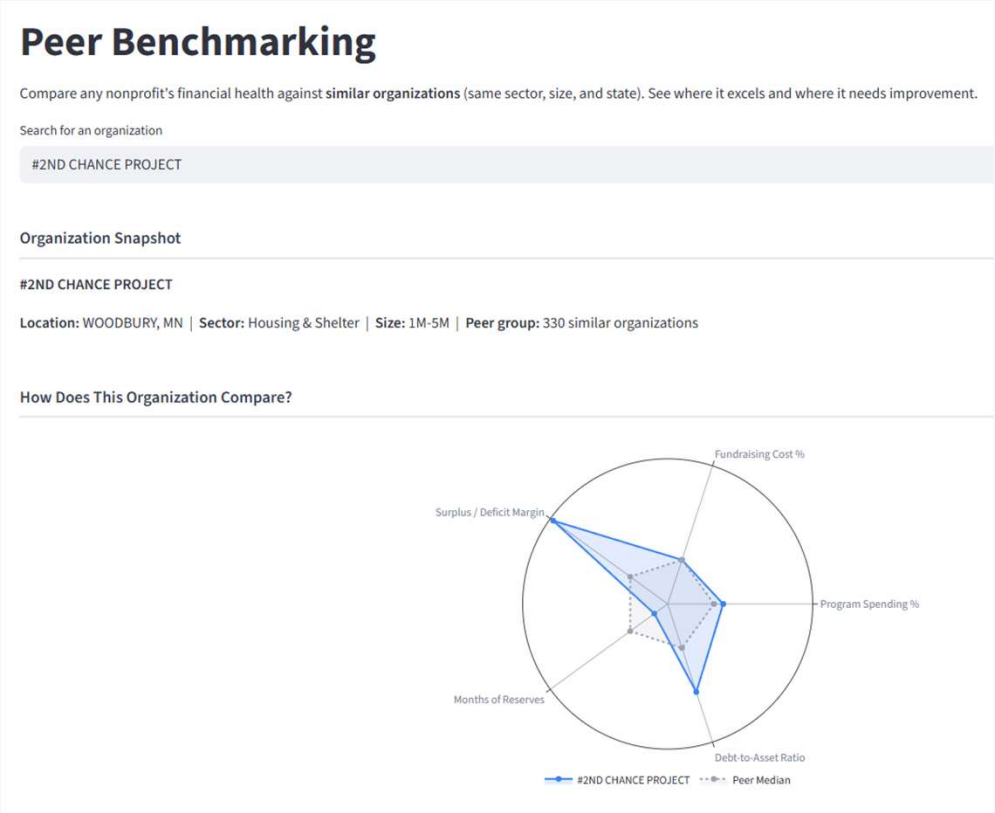
Peer Benchmarking

Why we built it

EDA finding #2 — sector and geography matter enormously. A raw score means nothing without knowing what 'normal' looks like for true peers.

What it shows

Radar chart of org vs. peer median across 5 axes. Green/red metric breakdown. Full peer group table with Z-score context.



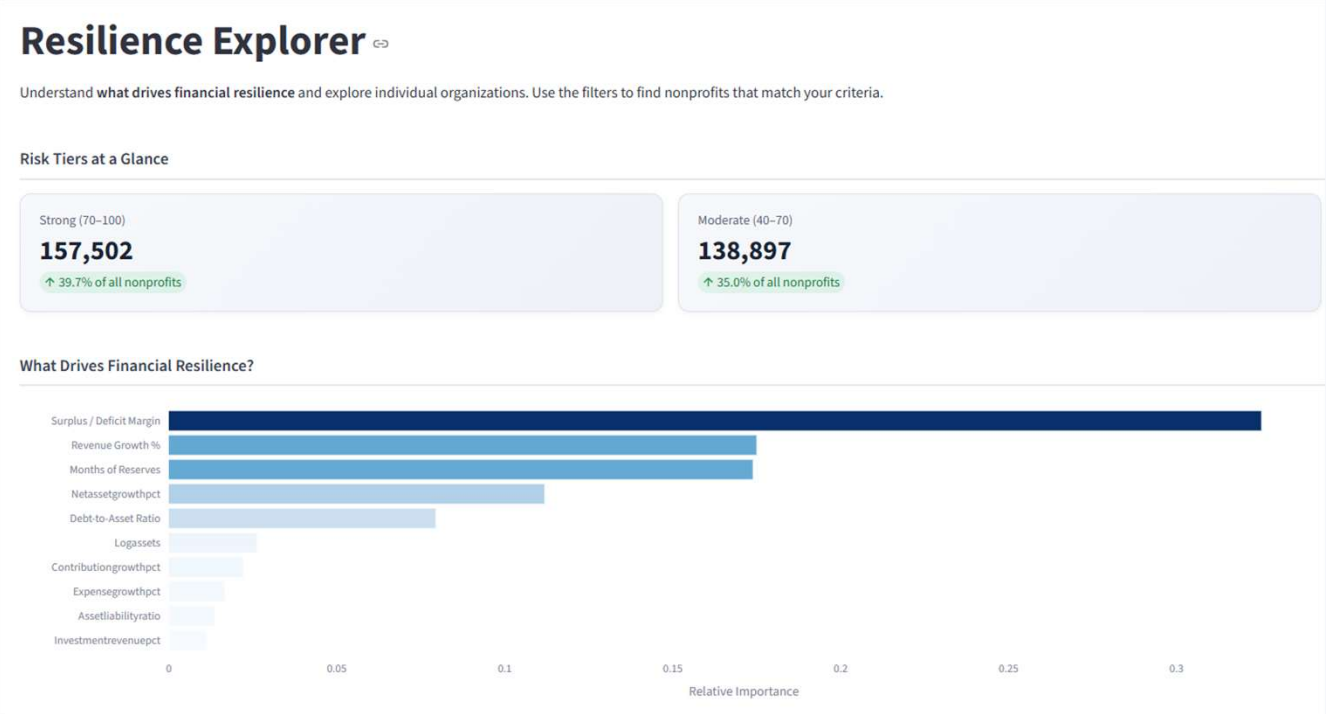
Resilience Explorer

Why we built it

Feature importance findings needed interactive surfacing — funders managing portfolios need ML risk probability and resilience score together.

What it shows

Feature importance bar chart from trained model, revenue vs. resilience scatter, filterable browse table with at-risk probability per org.



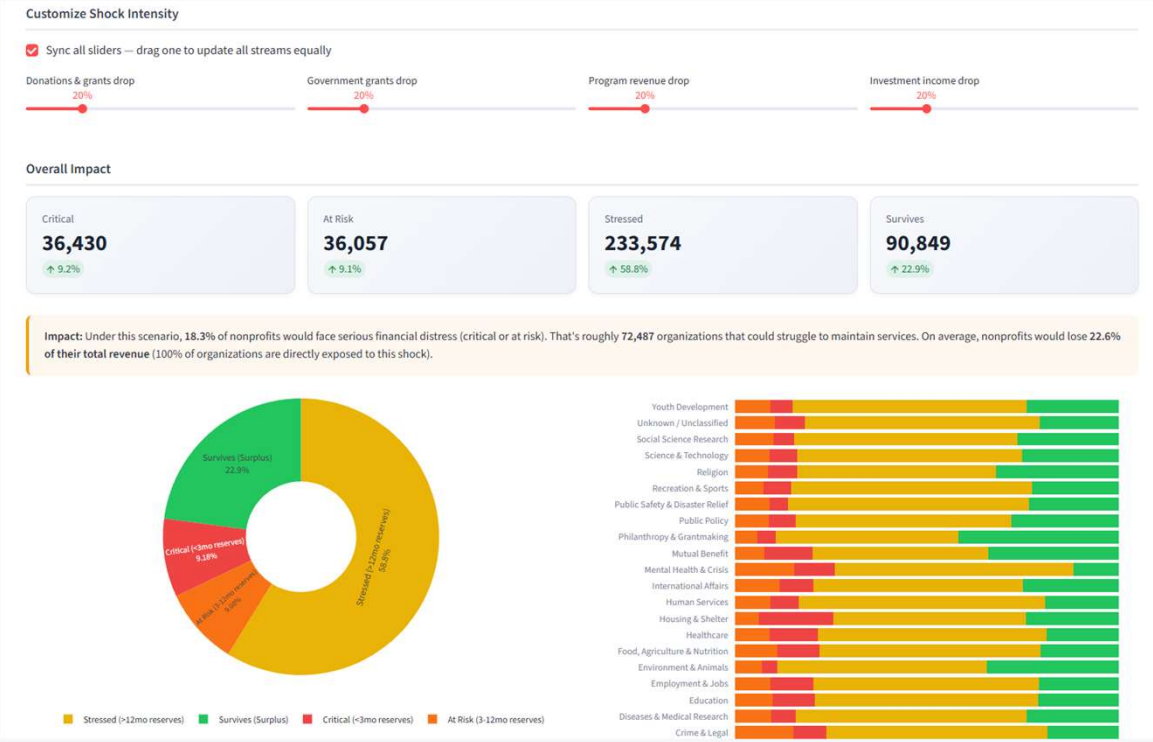
Stress Test Simulator

Why we built it

Knowing an org is 'at risk' today isn't enough — funders need to know what happens when a specific shock hits, especially given federal budget debates.

What it shows

5 pre-modeled scenarios, post-shock status breakdown (Critical / At Risk / Stressed / Survives), sector stacked bar, recovery timeline histogram.



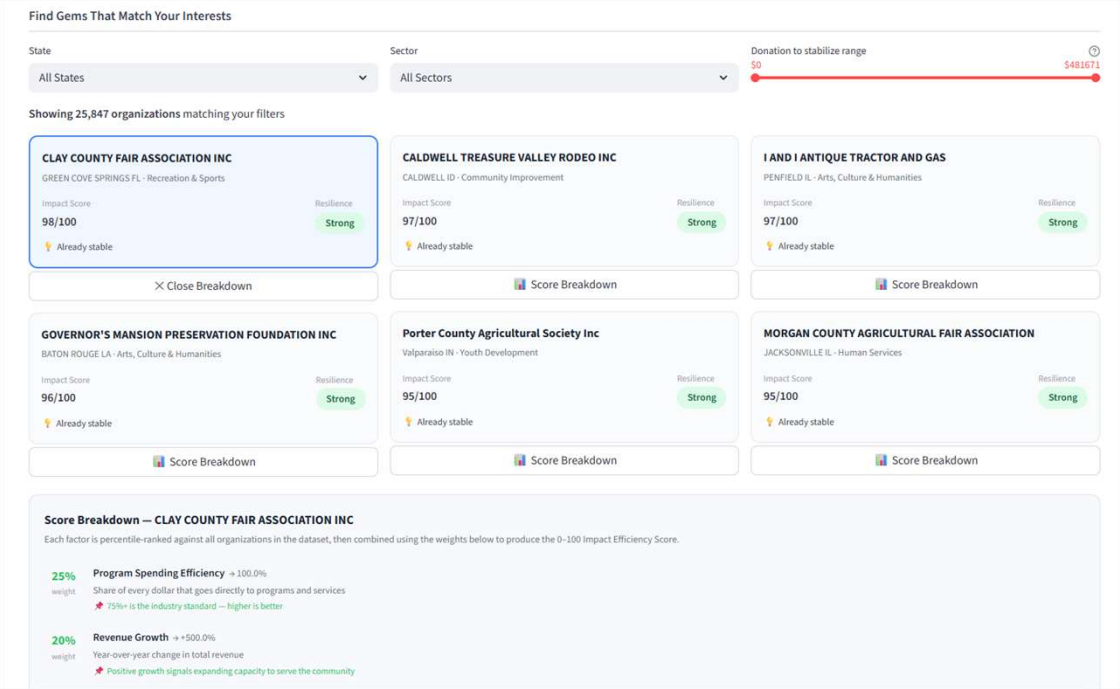
Hidden Gems Finder

Why we built it

Risk is only half the picture. EDA surfaced orgs punching above their weight — high impact efficiency, below-median budget, financially sustainable.

What it shows

27,208 qualifying orgs. DonationToStabilize metric — exact dollars to reach 6-month reserve. Top 6 gem cards. Budget vs. impact scatter.



Brand Map

Why we built it

The other tabs answer specific questions.
This one shows the entire sector at once — mission efficiency vs. financial survival.

What it shows

Program Expense Ratio (X) vs. Reserve Months (Y).
Dot color = resilience tier. Dot size = revenue.
Filterable by sector, state, size, year, tier.
Hover shows org name, revenue, surplus margin, grant dependency.

Quadrants

Low efficiency Well-resourced	High efficiency Well-resourced	High efficiency Fragile (hidden gems)	Low efficiency Fragile (highest risk)

Median: 79% program efficiency · 9.3 months of reserves